

PD-Intranet: Role-Based SharePoint Resource Center

Design Specification

Group 10

May 14, 2026

Contents

1 Overview

This document goes over every page and feature of the PD-Intranet project. It also lists all the packages and libraries needed to get it running in Docker.

2 Pages and Components

2.1 Login / Authentication

Users log in through Azure Active Directory. Once they are in, the system figures out their role and sends them to the right dashboard.

2.2 Role-Based Dashboard

This is the home page every user sees when they log in. What shows up depends on their role. We have 9 roles total: Attorney, Paralegal, Investigator, Social Worker, Admin Staff, IT, Management, Public Defender, and Intern.

2.3 Office Hoteling System

Allows staff to reserve desks in advance. Displays real-time desk availability per office location.

2.4 Attorney Workload Dashboard

Displays active case counts, upcoming deadlines, and workload summaries for attorneys and supervisors.

2.5 Urgency Portal

Powered by PowerBI. Displays analytics on high-priority cases and flagged items for management review.

2.6 Expert Witness Directory

A searchable directory of expert witnesses, filterable by specialty and availability.

2.7 Resource Libraries

Role-scoped document libraries. Staff only see folders and files relevant to their department.

2.8 Announcements

Office-wide and role-targeted announcements managed by admins.

2.9 CDD Resource Guide

A dedicated resource page for Capital Defense Division staff with documents and references specific to their role.

2.10 Admin / Maintenance Page

Available only to IT and admin roles. Allows management of SharePoint lists, role assignments, and content libraries.

3 Dependencies and Libraries

3.1 Frontend

Package	Purpose
SPFx (SharePoint Framework)	Web part framework for SharePoint
React	UI component library
TypeScript	Typed JavaScript for SPFx development
PnP.js	SharePoint REST API abstraction
Microsoft Graph API Client	Access Microsoft 365 data (users, groups)
SCSS / CSS	Styling for web parts

3.2 Backend / Data

Package	Purpose
Microsoft Graph API	Role and group resolution from Azure AD
Azure Active Directory	User authentication and permission sync
SharePoint REST API	Document library and list management
PowerBI Embedded	Analytics and urgency dashboards

3.3 DevOps / Deployment

Package	Purpose
Node.js	Runtime for SPFx build tools
Gulp	Build task runner for SPFx
npm	Package manager
Docker	Containerized deployment environment
SharePoint App Catalog	Deployment target for SPFx solution

4 Docker Container Setup

To run the development environment in Docker, the container requires:

- Base image: `node:18`
- Install global tools: `npm install -g gulp-cli @microsoft/generator-sharepoint yo`
- Copy project files and run `npm install`
- Expose port 4321 for the local SharePoint workbench
- Build with `gulp bundle` and package with `gulp package-solution`